

Understanding the Floor Experience: A Case Study of the Soft and Hard Floor Material Effect on Subjective Impression of the Exhibited Objects

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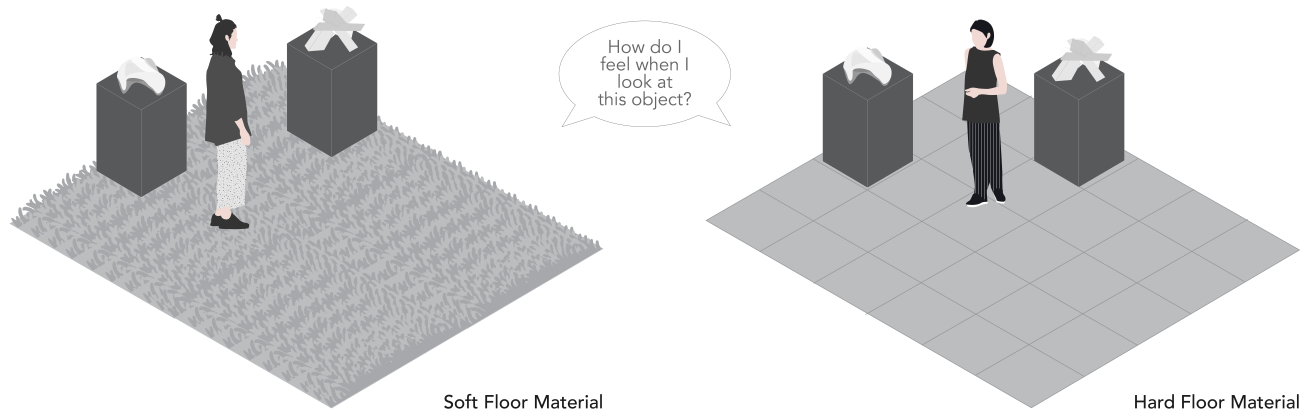


Figure 1: Experimental Setting and Research Question: Two rooms with the same dimension, layout, and objects but different floor materials (soft and hard). Can the floor materials interfere with the subjective impression during appreciation of exhibited objects?

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CHI EA '25, Yokohama, Japan

Abstract

Architecture and spatial design increasingly recognize the significance of subtle environmental factors that unconsciously influence

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ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3720112>

human behavior and cognition. While visual elements have traditionally dominated spatial design, the impact of tactile sensations as flooring materials on perception remains largely overlooked. This study examines how soft carpet and hard tile floors influence subjective evaluations of objects. Using abstract shapes inspired by the Bouba and Kiki phenomenon, 86 participants (44 on soft, 42 on hard flooring) evaluated two objects based on twenty-eight adjective pairs related to sight and touch within a controlled environment. Results showed that soft carpet floors enhanced perceptions of sophistication and beauty in Kiki, while hard tile floors amplified Bouba's elasticity and brightness. These findings highlight a "figure-ground relationship," where flooring texture interacts with sensory information to shape object perception. By focusing on the physical-material aspects of Human-Building Interaction (HBI), this research underscores the role of material effects in spatial design. It offers insights into how flooring can subtly guide cognition, behavior, and aesthetic experiences in architectural spaces.

CCS Concepts

• **General and reference** → **Empirical studies; Experimentation; Evaluation; Design.**

Keywords

floor material, subjective impression, material effect, object impression, human-building interaction

ACM Reference Format:

Burcu Nimet Dumlu, Kiryu Tsujita, Takatoshi Yoshida, Arata Horie, Tatsuya Saito, Yuh Yoshie, Kiyotaka Tani, Hisaaki Yokoyama, Keita Aono, and Kouta Minamizawa. 2025. Understanding the Floor Experience: A Case Study of the Soft and Hard Floor Material Effect on Subjective Impression of the Exhibited Objects. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720112>

1 Introduction

Architectural experience examines how the built environment shapes human perception through multisensory, cognitive, and emotional engagement. While architecture has incorporated multisensory approaches within the architectural atmosphere [41], the influence of subtle design features on perception and behavior is becoming increasingly evident. In this context, "subtle" refers to aspects of architectural space that are not consciously perceived as distinct objects but unconsciously affect human cognition and behavior.

Contemporary architectural design has primarily focused on creating visually distinctive and memorable experiences. However, recent research highlights the importance of incorporating auditory and tactile sensations into design. For example, in museums and galleries where artworks are displayed as visual objects, subtle cognitive experiences should also be considered. The spatial environment can shape how objects, including artworks, are perceived and experienced. Ideally, exhibition spaces should complement and enhance the artwork rather than overshadow or detract from it. On the other hand, in spatial designs that intentionally shape the perception of objects, such as those illustrated in Figure 2, it is essential to understand the conditions under which spatial design alters the impression of a given work. As Spence [36] pointed out,

"how atmospheric/ environmental multisensory cues interact" is a cross-modality problem of design practices.

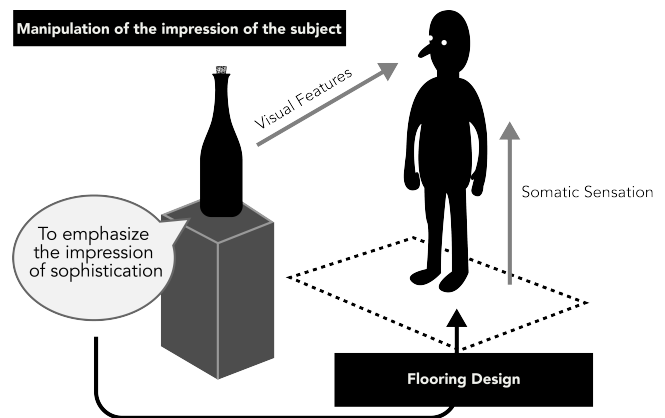


Figure 2: Subtle intervention in evaluating a subjective impression of exhibited object through the somatic textural design of flooring.

This study provides insights for multisensory material interaction. We conducted an experiment in two rooms with different floor materials—soft carpet and hard tiles—to investigate how these variations influence the subjective impression of objects placed within the space. We aimed to investigate the visuotactile interaction within the subtle material effect. By examining multisensory material interaction, we aim to contribute to the field of human-building interaction (HBI).

2 Related Literature

Scholars such as Norberg-Schulz [24], Pallasmaa [27], Tuan [39], Seamon [34], and Mallgrave [20] have extensively examined the relationship between space and human experience, emphasizing the role of sensory perception in shaping our understanding of the built environment. Factors such as materiality, sound, light, temperature, and spatial configurations play a crucial role in shaping human behavior and perception [26, 41]. Perspectives from phenomenology [34], cognitive science [37], and neuroscience [20] contribute to understanding how spaces are perceived and experienced through embodied interaction [9]. Dourish [9] introduced the concept of embodied interaction, arguing that interaction is not merely about abstract logic or digital representations but about how people engage with technology in real-world, social, and material contexts. This perspective aligns with the idea that human experience is deeply embedded in the physical and sensory aspects of the environment within and without technological aspects. This paper emphasizes material contexts while highlighting their potential for future technological advancements in HBI.

HBI is an interdisciplinary field that explores the interplay between human experiences and the built environment. It intersects Architecture and Urban Design, Ubiquitous Computing, and Interaction Design, synthesizing knowledge from each field and offering valuable insights into all of them [2]. As Becerik-Gerber [4] mentioned, the goal of HBI is to examine people as individuals

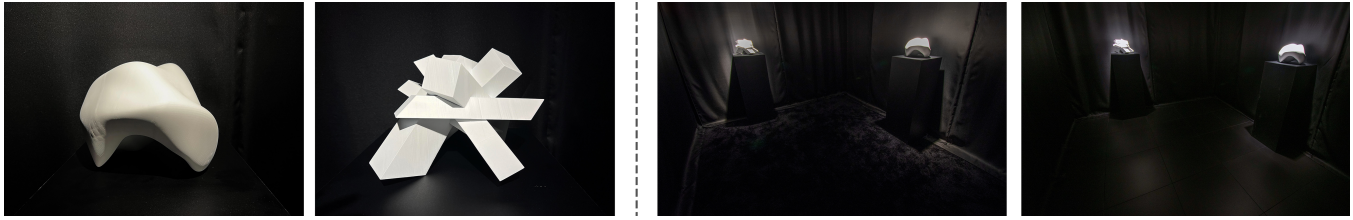


Figure 3: Exhibited Objects in the Experiment: (left) Bouba (1), Kiki (2), Rooms (Right) Soft Floor (1), Hard Floor (2)

with distinct life experiences, personalities, abilities, and objectives rather than just inhabitants, end users, or uniform agents in a built environment. These different people engage in activities within and around the built environments where they live, work, and play, HBI research seeks to promote good engagements [4]. Becerik-Gerber et al. [4] defines "the research dimensions of HBI as humans, buildings, and technology." Their HBI framework envisions a future in which buildings become perceptual and cognitive environments, involving both users and the physical infrastructure that influences and supports human purpose, perception, and behavior [4]. Alavi et al. [3] conceptualized that "the HBI within three interconnected aspects 1) physical-material, 2) spatial-configurational, and 3) social-cultural". In this study, we aim to contribute to the physical-material dimension of HBI.

Economidou et al. [10] explained HBI within the framework of lived experience, referencing Merleau-Ponty [21] and other scholars, define human experience as inherently "sensory," shaped by architectural and spatial design. Research studies have shown that spatial designs incorporating multiple sensory elements can subtly affect cognition and behavior. As Spence [36] explained, environmental cues significantly affect humans even though they are unaware of it. As a visual example, Okken et al. [25] showed that the number of utterances a person makes changes depending on the size of the room and the desk. Meyers et al. [22] revealed that ceiling height affects thinking and task performance. Another study by Meyers-Levy et al. [23] suggested that bodily sensations can shape the product perceptions and the contrast effect of product viewing occurs within close distances. In olfaction, Liljenquist et al. [18] showed that people move more virtuously in a detergent-scented room than in an odorless room. In a study of subtle influence in tactile sensation, Ackerman et al. [1] investigated the effects of chair hardness and blanket softness on social judgment and decision-making.

While many elements contribute to architectural design, this study focuses specifically on the floor. Gibson [12] claims that while the floor alone provides a platform for standing, walking, or running, it is essential not to isolate it from the real-world context. By inhabiting, it can support the diverse range of activities associated with everyday life [12, 13]. Ingold [15] explains that the significance of the floor lies in its relationship with other elements, whether it is a space for interaction, movement, or perception. It is a surface embedded in a web of relations, not an isolated backdrop. As floor surfaces serve as physical contact points between people and architecture, incorporating tactile indicators and evaluations into the design is essential. However, floor design is often assessed primarily based on visual aspects, such as the appearance and material

texture. Some studies have explored the effects of flooring materials on individuals, including evaluations of subjective discomfort and fatigue when standing on various flooring surfaces [6, 29]. Another study by Meyers-Levy et al. [23] resulted that standing on soft carpet induced sensation of comfort while the hard tile caused discomfort or fatigue. Also, material effect is an investigation point for psychology research. When stomping on textures, testing them with the sole of the foot or the hand, Cleland et al. [7] demonstrates that roughness, hardness, and stickiness ratings are often consistent and perceived stability is also influenced by hardness.

Also, the floor has different investigation points for Human-Computer Interaction and HBI areas; interactive floor [28], digital floor [8], and tactile floor [14, 38, 40]. Spatial experience in Human-Computer Interaction has been widely investigated by different scholars [5, 31]. Haptic and tactile floors are growing areas within the HCI and HBI.

After all the studies mentioned regarding floor experience, this study aims to provide insights into the interaction of soft and hard materials' tactility in the context of exhibited objects. We aimed to clarify whether tactile textures (soft and hard) influence the impression of objects, such as artworks, beyond their impact on personal sensations. This study analyzes the subtle tactile effects of floor surface textures (soft and hard) on people's perceptions by having participants evaluate objects. The experiment investigates the subjective impressions that may occur when objects are appreciated in spaces differentiated by various floor surface materials.

This study employed an experimental setup to investigate how flooring materials influence the perception of objects. Participants evaluated two abstract shapes under two flooring conditions: soft carpet and hard tiles (Figure 1). We aimed to contribute to HBI by understanding the material effect on cognition and behavior.

3 Experiment Design and Method

We designed an experiment to explore the differences in human somatic experiences when interacting with soft versus hard floors while observing two objects in a room. Two rooms had identical features, including covered walls and spotlights highlighting the objects with the same angle and brightness. The sole difference between the rooms was the floor material: one room had a soft carpet, and the other featured hard tiles, as shown in Figure 3 (right). In each room, two 3D objects were placed: one inspired by the Bouba motif and the other by the Kiki motif, based on research into the associations between shapes and various impressions [11, 16, 32] as illustrated in Figure 3 (left). The room settings and object placements were treated as dependent variables, while the floor material was an independent variable and the control point.

The experiment procedure required the participants to enter a room with an iPad and complete an online questionnaire containing sensory evaluation questions about object assessments. The participants were not informed that the experiment aimed to examine the effect of floor material on their subjective impressions of the objects. Therefore, we did not ask them to change their shoes; they entered the room wearing their own. After entering the room, they responded to the sensory evaluation questions for both Bouba-based and Kiki-based objects using a 7-point Likert scale and left the room.

The sensory evaluation scale was developed based on a study by Shimizu et al. [35], which identified adjectives associated with the senses of sight and touch. It included twenty-eight adjective pairs: like-dislike, beautiful- dirty, noisy- quiet, flashy- plain, light-heavy, bright- dark, sharp- dull, good- bad, dynamic- static, cheerful-gloomy, tense- relaxed, sturdy-fragile, warm- cold, simple- complex, strong impression- weak impression, sophisticated- tacky, calm-anxious, comfortable- uncomfortable, rough- smooth, elastic- inelastic, regular- irregular, modern- archaic, luxury- cheap, unique-typical, refreshing- annoying, I want to touch it- I don't want to touch it, a close presence- a distant presence, and I want to keep looking- I don't want to keep looking [35].

The scale was structured on a 7-point system, with responses ranging from 1 (extremely) to 7 (extremely) via intermediary options: 2 (very much), 3 (somewhat), 4 (neither), 5 (somewhat), and 6 (very much). Participants evaluated the adjective pairs using this framework. We analyzed the data collected from two rooms, identified key patterns in perception, and examined how different flooring materials influenced participants' sensory experiences. Based on these findings, we proposed an internal model for integrating visual and somatic sensations, highlighting the interplay between material properties and subjective impressions.

3.1 Participants

86 participants, aged between their 20s and 50s (45 males and 41 females), joined the study. Each was assigned to one of two rooms, with 44 participants entering the soft-carpeted room and 42 experiencing the hard-tiled room. The difference in group sizes resulted from the unexpected cancellation of two participants originally assigned to the hard-tiled room.

Before entering, participants were briefed on the experiment without being informed that it specifically related to the floor. Each participant received an iPad to complete the questionnaire. Upon entering the room, they were instructed to observe the two objects for approximately two minutes. Afterward, they proceeded to the next step and completed the questionnaire, rating both objects on a 7-point scale for all 28 adjective pairs. The experiment ended after the participants completed the questionnaire and left the room.

3.2 Experimental Apparatus

Two rooms were prepared for the experiment: one with a soft carpet floor (Hantafu Black W100, 30 mm shaggy, 1800 grains, custom-made for Nomura Co., Ltd.) and the other with hard tiles (Sanwa Co. MB-6000M, matte finish [33]) (see Figure 4). Each room measured approximately twelve square meters (3.96 m × 2.875 m), with black as the predominant color to minimize visual distractions.

Soft carpet and hard tiles contrast in material compliance and resistance, influencing physical interaction and perceptual experience. Carpet, composed of fibrous materials with a cushioned underlayer, exhibits high deformability and effective pressure distribution, allowing it to yield under force. This quality absorbs mechanical impact, reducing stress on the body and creating a sensation of tactile continuity.

Rys and Konz [30] found that after two hours of standing, heart rate (beats per minute) was higher on a concrete floor (11.1 bpm) compared to carpet (95.2 bpm). Additionally, perceived comfort, measured on a 100-point scale, was greater when standing on carpet (mean 77.7) than on concrete (mean 71.9) [30]. The compressibility of carpet minimizes immediate haptic feedback, fostering a sensation of softness, warmth, and spatial envelopment. As a result, carpeted surfaces create a sense of embeddedness, influencing movement patterns and proprioceptive awareness by modulating force perception.

Madeleine et al. [19] reported that prolonged standing on a soft surface resulted in lower levels of unpleasantness and greater comfort compared to a hard surface. Standing on hard flooring led to increased shank swelling, higher electromyography (EMG) activity in the right soleus muscle, greater muscle fatigue, and negative changes in standing posture in three participants [19]. While postural activity was reduced on the soft surface, it remained sufficient to prevent lower leg swelling [19]. These insights into material effects and their impact on the body motivated us to use soft carpet and hard tiles.



Figure 4: The soft (left) and hard (right [33]) materials applied to the experiment rooms

After creating the contrast material effect, we designed the two rooms for participants to create the minimum visual difference. We placed the spotlights directly on the objects and created directed, dimmed lighting. The aim was to focus the participants' attention on the objects rather than the floor. As Figure 3 (right) shows, the floor has a slight visual difference.

The evaluation objects were custom-designed and 3D-printed by the authors to represent Bouba and Kiki, commonly used in cross-modal perceptual evaluations. Each object was placed on a 1.1-meter-high platform (Figure 3 (left)). Bouba and Kiki were used as a means of experimenting with multisensory integration. Bouba and Kiki were chosen to investigate multisensory integration, as previous research by Watanabe et al. [32] demonstrated that the Bouba/Kiki effect extends to tactile sensations.

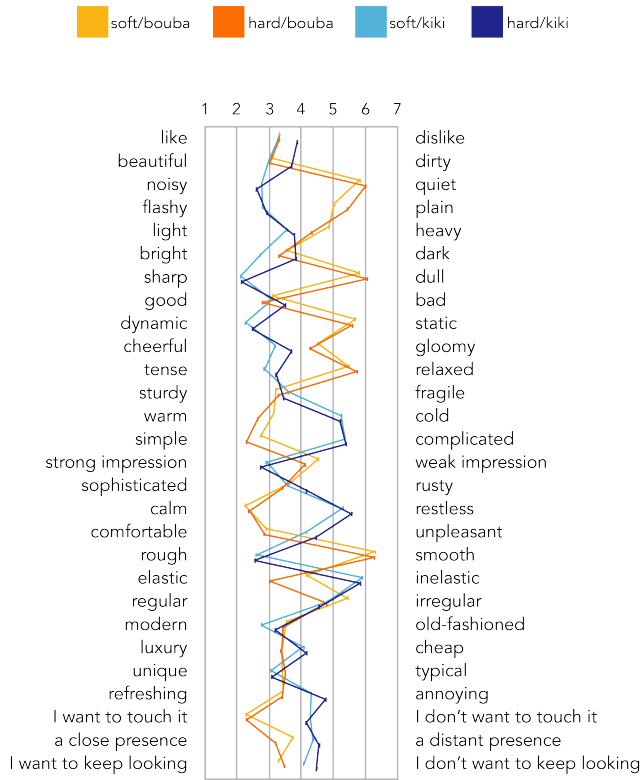


Figure 5: The results from the 28 pairs of adjectives questionnaire

We collected both the questionnaire data and video recordings from each room. Two GoPro cameras were installed on the ceiling of each room to record and verify participants' behavioral experiences. We obtained their consent and formalized the consent in the questionnaire.

3.3 Data Analysis

The subjective evaluations of the objects (Bouba and Kiki) were analyzed by using the answers for twenty-eight adjective pairs within 7-point Likert scale, with mean ratings recorded for each room and object (see Figure 5). The analysis focused on identifying differences in perception based on flooring materials (soft vs. hard floors).

A t-test for independent samples was conducted for each adjective pair to determine significant differences in ratings across the two flooring conditions.

The results are categorized by object: significant differences ($p < 0.05$), trends ($p < 0.1$), and non-significant results. These findings are visually summarized in Figures 5, 6, 7, and 8. Figure 5 shows all adjective pairs for both rooms and both objects. Figure 6 shows the significant results for the Bouba object in the soft carpet and the hard tile room. Figure 7 illustrates the significant results for Kiki object in both rooms. Lastly, Figure 8 provides an example of the non-significant results for soft carpet and hard tile experiences.

3.4 Findings

We analyzed 86 participants' data for each object and revealed the findings below.

3.4.1 Bouba. For Bouba, a significant difference ($p < 0.01$) was observed in the **elastic/non-elastic** adjective pair, suggesting that the flooring texture influenced perceptions of elasticity (see Figure 6). Participants perceived Bouba as more elastic on a hard tile floor compared to the soft carpet floor. Additionally, a significant trend ($p < 0.1$) emerged for the adjective pair **regular/irregular**, indicating a tendency for Bouba to feel more irregular on a soft floor.

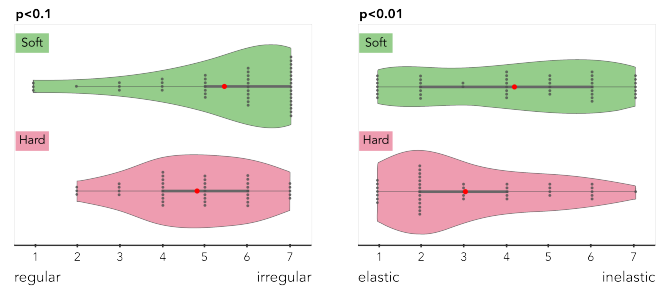


Figure 6: Results for Bouba object: regular/irregular and elastic/inelastic

3.4.2 Kiki. In Kiki, three adjective pairs showed significant differences (see Figure 7): **Liked/disliked** ($p < 0.05$): Participants expressed a preference for Kiki in the soft floor condition. **Bright/dark** ($p < 0.001$): Kiki was perceived as lighter on the soft floor compared to the hard floor. **Beautiful/dirty** ($p < 0.01$): Kiki was considered more beautiful when displayed on a soft floor.

A significant trend ($p < 0.1$) was also noted for the adjective pair **sophisticated/tacky**, indicating that Kiki tended to feel more sophisticated on the soft floor.

3.4.3 Example of Statistically Non-Significant Results. Adjective pairs that did not show significant differences between the two flooring conditions are summarized in Figure 8. These results suggest that not all sensory perceptions are equally influenced by floor material, underscoring the complexity of the interaction between tactile and visual elements in object evaluation.

4 Discussion

The results demonstrate that different flooring types can influence subjective evaluations of certain specific properties of an object. Each adjective pair is discussed below:

The **elasticity** of Bouba was perceived more strongly on a hard floor. The softness felt underfoot on a soft floor may influence visual perception, diminishing the perceived elasticity of the object. Conversely, a hard floor may enhance the perception of elasticity. Since Kiki was less likely to be perceived as elastic, this difference was considered negligible.

Regarding **irregularity**, participants perceived Bouba as more irregular on a soft floor, whereas Kiki was perceived as more irregular on a hard floor. The tactile feedback from the soft floor may reduce

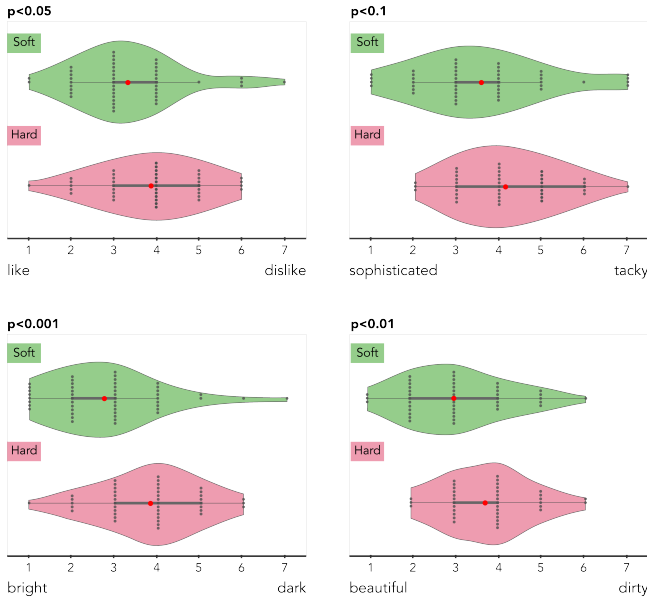


Figure 7: Results for Kiki object: like/dislike, sophisticated/tacky, bright/dark, and beautiful/dirty

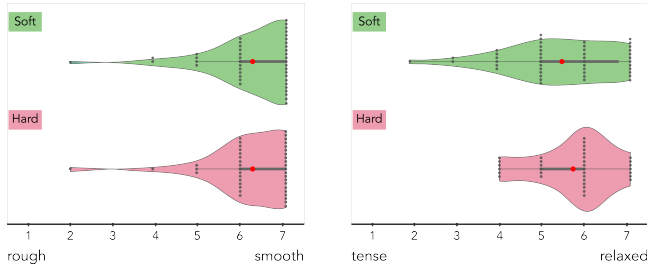


Figure 8: Example results with the statistically non-significant: rough/smooth and tense/relaxed

the perception of regularity through the soles of the feet, potentially influencing the visual perception of the object's regularity as well. For **brightness**, Kiki was perceived as brighter on the soft floor. The softness of the floor may have provided a more comfortable tactile experience, subtly enhancing the perceived whiteness of the object. Alternatively, the soft floor may have absorbed more light, darkening the surrounding environment and increasing the object's relative luminance.

In terms of **aesthetics**, participants tended to perceive Kiki as more beautiful on softer floors. The tactile impression of softness and irregularity from the floor contrasted with Kiki's linear form, emphasizing its visual sharpness and enhancing its perceived beauty. In Bouba's case, the object's inherent beauty appeared to overshadow any noticeable influence from the flooring.

When considering **liking and disliking**, participants tended to favor Kiki on softer floors. The tactile softness and irregularity of the soft floor seemed to accentuate Kiki's linear visual features, positively influencing the overall impression of the object.

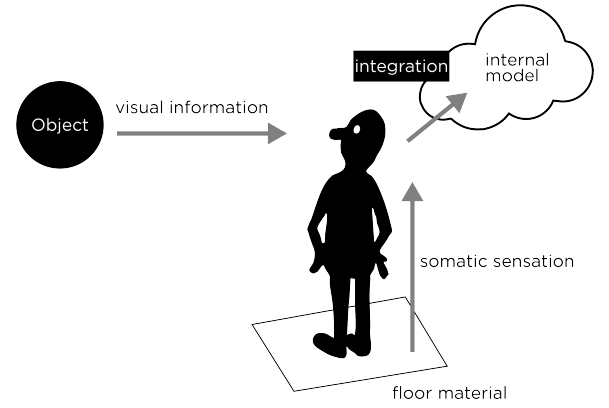


Figure 9: Hypothesis of the internal model on the integration of visual and somatic sensation

Similarly, participants perceived softer floors as more **sophisticated** in relation to Kiki. The irregular tactile sensation of the soft floor may have heightened the object's relatively linear characteristics, making it appear more refined.

For adjectives where no significant differences were observed (e.g., rough/smooth, tense/loose), the variation in distribution may still influence subjective evaluations, warranting further investigation. These findings suggest that the results can be understood through a "figure-ground relationship," a perceptual phenomenon that examines the interplay between a target object (Figure 9) and its surrounding environment (ground). In this experiment, the flooring material functioned as the ground, influencing evaluations of the object.

The texture of the flooring appears to influence object evaluation, either consciously or unconsciously. After analyzing the experiment results, we proposed an updated version of Figure 2—an internal model for integrating visual and somatic sensations (see Figure 9). We proposed that two patterns of integration may occur; additive and contrastive integration. **Additive Integration**, where similar sensory qualities are mutually emphasized between the object and flooring. **Contrastive Integration**, where contrasting sensory qualities highlight the object's features.

In this model, visual information about the object is combined with somatosensory input from the flooring, which is likely processed through an internal cognitive framework.

5 Conclusion

This experiment examined the subjective impressions of the displayed object perception within spaces differentiated by flooring materials (soft carpet and hard tiles). Two objects were used, inspired by the Bouba-Kiki phenomenon: one with a smooth, rounded shape (*Bouba*) and another with a jagged, spiky shape (*Kiki*). Although our aim was not to investigate the sound-shape relationship, this phenomenon helped guide our exploration of shape perception and how flooring materials influence it. Therefore, we designed the Bouba and Kiki objects, 3D-printed them, and placed them in both rooms.

Participants entered only one of the rooms wearing their own shoes. They used an iPad to complete an online questionnaire about the objects. The questionnaire consisted of 28 pairs of adjectives rated on a 7-point Likert scale, selected based on the object they were observing. They provided responses for both objects.

Participants answered the questionnaire about the objects in controlled environments designed to isolate the influence of flooring texture. The results revealed that flooring textures significantly affected certain perceptions, with soft floor enhancing impressions of preference, sophistication, beauty, and brightness in Kiki. In contrast, hard floor amplified perception of elasticity in Bouba. Also, these findings highlight the subtle yet impactful role of tactile flooring elements in shaping visual and cognitive evaluations of objects, underscoring the importance of integrating multisensory approaches into spatial design. According to these results, we proposed an internal model on the integration of visual and somatic sensation. All the findings and proposed internal model offer practical guidelines for flooring design in the context of object perception in museums and gallery spaces.

As the next step, we will conduct an experiment to examine how people perceive paintings in the same setting, comparing the effects of soft carpet and hard tile flooring. In the third iteration, we will investigate the effects of a stress condition on the decision to leave the room with the same materials applied to the rooms as in experiments one and two. This series of studies focuses on floor-material interaction. Lawson [17] defines architecture as a sensory experience shaped by touch, sight, and sound, emphasizing its role as a form of language that communicates meaning beyond aesthetics. This supports the idea that material choices in architecture are not merely technical decisions but have significant cognitive and emotional impacts on users.

While flooring has traditionally been a static, one-time choice, recent advances in sensing technologies such as load cells integrated into building materials and vibration actuators in floors create new possibilities for dynamically influencing perception and behavior. Integrating these technologies could enable floor designs that achieve more sophisticated effects in future applications. After conducting the experiments and applying new sensing technologies, our main goal is to create an extensive material catalog that provides insights for architecture, HCI, and HBI.

Acknowledgments

The authors sincerely express their gratitude to all the participants who took part in our study and generously shared their data. The Ethics Committee of the Keio University Graduate School of Media Design has approved this experiment (#2023-19-3).

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